Hitelecom®

EXPLOSION-PROOF TVOC SENSOR

H Series | Industrial TVOC Monitoring

H200 LoRa | H300 4G | H500 NB



KEY INDICATORS

Measuring Range	Resolution	Wireless	Protection	Sampling
0-100,000 ppb 0-100 ppm	1 ppb Reported increment	4G / NB-IoT / LoRa	IP68 option* Enclosure by order	Customer-defined Separate reporting

OVERVIEW

The Hitelecom H Series TVOC sensor combines a configured VOC sensing element with a wireless terminal for solvent-handling and industrial process areas. Select electrochemical or PID sensing for the target compounds, with a published range of 0-100,000 ppb (0-100 ppm) and 1 ppb resolution. Data collection is customer-configurable; reporting is set separately. Match H200 LoRa, H300 4G or H500 NB-IoT, the external 9-24 V supply, mounting and platform integration to the project. Explosion-proof configuration, Ex marking and certificate scope follow the supplied specification.

FEATURES

CLOUD

Send configured VOC readings to Hitelecom Cloud or a compatible customer / private platform.

COMPOUND-LED

Electrochemical or PID sensing, by selected configuration. Match response to target compounds.

YOUR INTERVAL

Customer-defined data collection. Set reporting separately around process and monitoring requirements.

EXPLOSION-PROOF

Match the ordered Ex marking, certificate and installation limits to the classified location.

EXTERNAL POWER

External 9-24 V power supply. Match the power wiring to the ordered installation requirements.

NFC / ALERTS

NFC activation and local setup. Configured threshold alerts through the agreed platform workflow.

APPLICATIONS


Coatings & printing


Packaging & adhesives


Tire & rubber


Chemicals & storage


Petrochemical handling


Pharma & laboratories


VOC treatment & recovery

Illustrative VOC-process environments; define monitoring points for the solvent mixture, airflow and installation.

Item		Description
Physical Characteristics	Product models	H200 (LoRa) / H300 (4G) / H500 (NB-IoT)
	Power supply	External power supply
	Input voltage	9-24 V
	Display / alarm hardware	Display and local alarm hardware follow the ordered configuration.
	Mounting	Ear mount / duct mount, by selected installation arrangement.
	Dimensions	Terminal, sensing head and accessory dimensions follow the supplied configuration.
Connectivity	Cellular data protocol	MQTT for the cellular configuration; agree receiving-platform requirements.
	LoRa data path	Match the supplied radio protocol, receiver / gateway and onward data route.
	LTE band reference	TDD B34 / B38 / B39 / B40 / B41; FDD B1 / B3 / B5 / B8 (LTE reference).
TVOC Measurement	Measuring range	0-100,000 ppb (0-100 ppm)
	Resolution	1 ppb (reported increment)
	Sensing principle	Electrochemical or PID, by selected configuration.
	Measurement performance	Confirm accuracy, detection limit and response time for the selected sensing configuration and target compounds.
	Data collection interval	Customer-configurable; confirm the supported settings for the selected configuration.

Sampling and reporting are separate settings. Match the target compounds, calibration basis, power and update timing to the monitoring task.

SPECIFICATIONS

Item		Description
Waterproof Rating	Enclosure option	IP68 customization is listed; the supplied assembly's rating follows the order.
	Sensing / entry protection	Confirm protection for the sensing head, inlet, cable entry and accessories.
Reliability	Calibration / checks	Agree the calibration basis and function-check schedule for the sensing element.
	Maintenance	Follow supplied instructions; keep calibration and service records for each point.
Environmental Range	Operating temperature	Follow the sensing-element limits and the certified assembly's ambient limits.
	Humidity / exposure	Confirm moisture, condensation, dust and chemical-compatibility limits.
	Storage conditions	Follow the supplied TVOC assembly's storage conditions.
Integration	Data destination	Hitelecom Cloud or a compatible private platform; cellular integration uses MQTT.
	LoRa integration	Agree the protocol, regional channel plan, receiver / gateway and server route.
	Reporting / alerts	Set reporting separately from collection; agree thresholds and response workflow.
Explosion-proof Configuration	Ex marking / certificate	Match the ordered Ex marking, certificate scope and special conditions to the installation.
	Area requirements	Specify area classification, gas / dust group and required temperature class.
	Complete assembly	Review sensor, enclosure, power, cable entries and alarm accessories for the required approval.
Certification References CCC / CE / FCC / RoHS	CE platform status	H200 / H300 / H500 CE completed; common wireless-platform scope.
	Ex / other documents	Confirm explosion-protection and CCC / FCC / RoHS documents for the supplied configuration.

* 1 ppb is resolution. Accuracy, detection limit and response time follow the target compounds and sensing configuration.

* External 9-24 V power supply. Data collection is customer-defined; the reporting cadence is configured separately.

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